

RB14 replay test guide

Record these as BeamNG replays (.rpl) and add them to the repo: tools/rb14/replay_analysis.py reads them.

For every recording

- **Stop for 3 s between tests** - the analysis splits the replay at standstills.
- **Clean inputs:** throttle fully on or off, brake fully on, steering held steady. Partial inputs blur the numbers.
- **Both directions** for every cornering test (left, then right).
- **Hands off the steering on straights** after a turn: centre it and leave it - a small correction hides the drift we are measuring.
- **Write down** the car configuration and controller (wheel / pad / keyboard). Keep each replay to ~2-3 min (one replay per block below).
- Big flat area (gridmap is fine); the longest straight you have for the top-speed runs (3 km+).

Replay A: straight-line tests

#	What to do	What it tells us	Done
1	Spawn and wait 10 s without touching anything.	Spawns stably? Real wheel alignment at rest (it showed the 3.7° rear toe), ride height.	
2	Standing start, full throttle , steering centred, through the gears until the speed stops rising.	0-100/200/300 km/h, acceleration at each speed, real top speed, gear changes, traction control.	
3	Then lift off and coast - no brake, no throttle - down to ~100 km/h.	Most useful part. Coast-down separates air drag (grows with speed ²) from rolling / scrub losses (constant): pins down the ~3 kN of extra drag.	
4	Accelerate to ~300 km/h, then full brake to a stop, steering centred.	Braking g at each speed, brake balance, lock-ups, pull under braking.	
5	Repeat 2 with DRS open.	DRS gain at top speed.	
6	Hold a steady ~150 km/h for 10 s , steering centred.	Does it track straight before any cornering? (reference for the drift)	

Replay B: low-speed cornering (~60-100 km/h)

#	What to do	What it tells us	Done
7	Steady circle, left. Fixed steering angle, raise speed slowly over ~15 s until it slides. Then straighten, centre the wheel, drive straight 5 s .	Low-speed grip limit, understeer / oversteer, drift after the corner (current complaint).	
8	The same, right.	Left / right comparison.	
9	Hairpin, left: brake hard into a tight turn, full throttle on the exit, then 5 s straight with the steering centred.	Traction out of slow corners, wheel judder, rear-axle steer, recentring.	
10	Hairpin, right.	The same, other side.	

Replay C: high-speed cornering (~180-250 km/h)

#	What to do	What it tells us	Done
11	Long fast sweep, left , steady steering ~5 s at high speed, then 5 s straight with the steering centred.	High-speed grip with downforce, front-wheel wobble, front-wing movement, ride height under load.	
12	The same, right.	Left / right comparison.	
13	Quick lane changes at ~150 km/h: sharp left-right-left flick, then hold straight.	Steering response and damping; do the wheels or wing wobble after a quick input?	

Car / configuration: _____ Controller: _____ Map: _____
